DO NOW — RECALL

Lesson 046 · Objects in Arrays and Lists

Name: _____ Class: _____ Date: _____

1. What is a constructor used for?

2. What does private mean for an instance field?

3. What index identifies the first C# array item?

4. A rising-edge D-type flip-flop starts with Q=0. D=1 at the first rising edge. D then falls to 0 before the falling edge and stays 0. Trace Q immediately after the first rising edge, the falling edge and the next rising edge.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Initialising an object's state.

2. Model answer

It cannot be accessed directly from outside the class.

3. Model answer

Zero.

4. Model answer

1, 1, 0. Q samples D only at each rising edge and holds its value between those edges.

By the end, you can…

• A class describes an object; the array stores references to two Book instances.

• Indexes start at 0. shelf[1] identifies the second reference, then .Title reads its property.

• An array has a fixed length after creation; individual array elements can refer to objects whose properties change.

Vocabulary — know these terms

object — Instance of a class.

reference — Value identifying an object instance.

aliasing — Several references to the same object.

index — Position in a sequence.

NEW LEARNING

Read, annotate and explain

Observe

A class describes an object; the array stores references to two Book instances.

Explain

Indexes start at 0. shelf[1] identifies the second reference, then .Title reads its property.

Apply

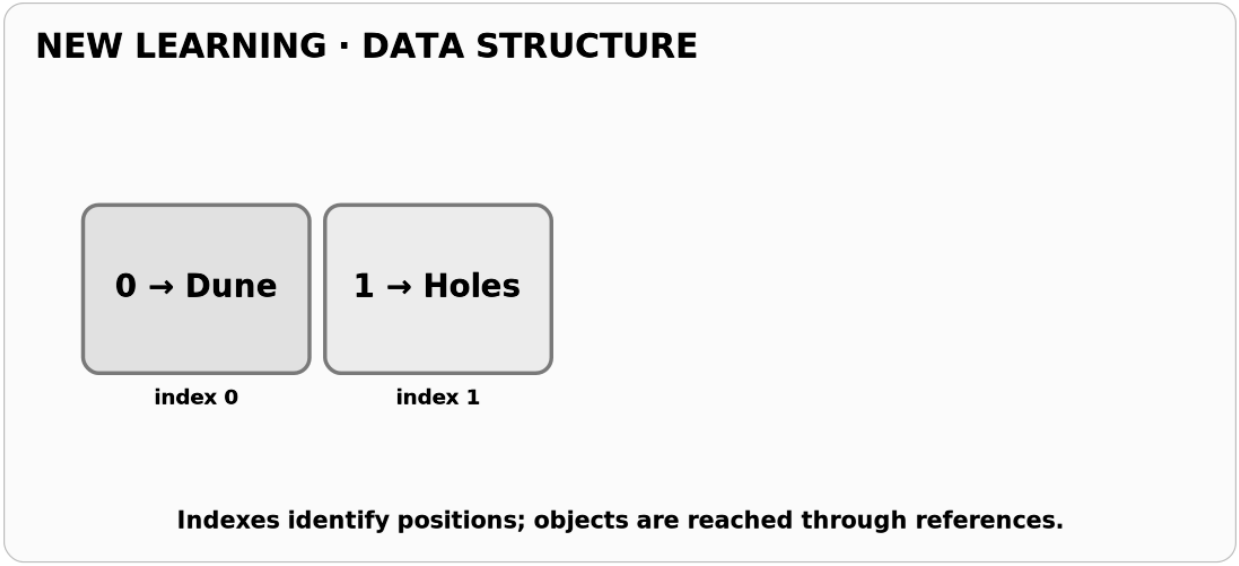
An array has a fixed length after creation; individual array elements can refer to objects whose properties change.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



NEW LEARNING · DATA STRUCTURE: 0 → Dune, 1 → Holes

I DO — WORKED EXAMPLE

Follow the worked method

Trace the title stored at books[0] in a two-object Book array.

Trace and justify

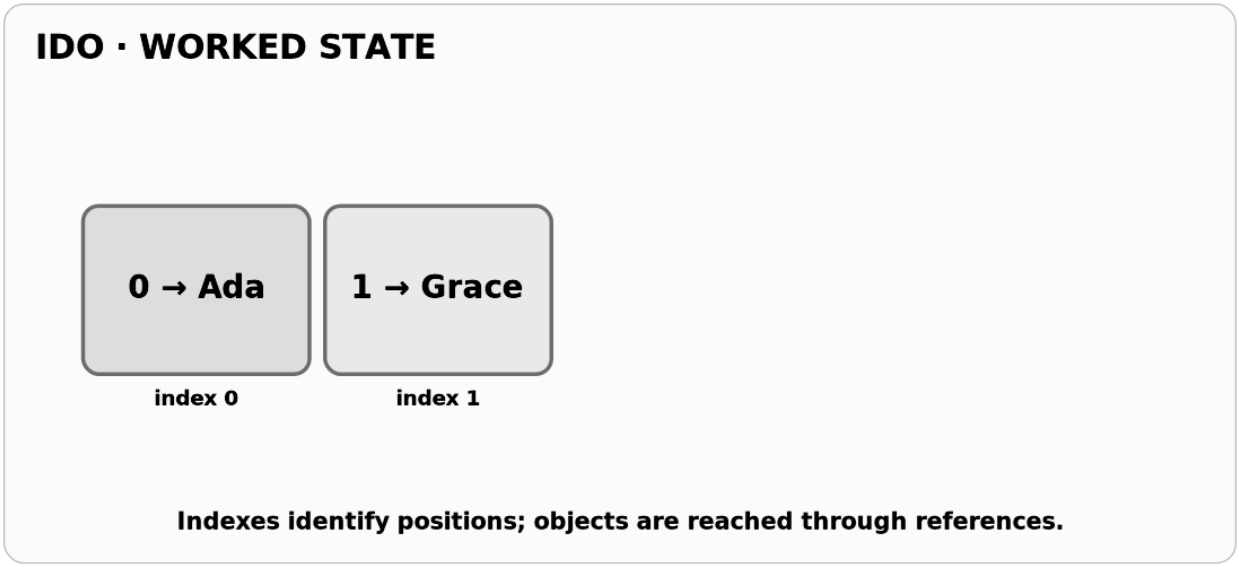
Index 0 refers to the first object; its Title is Ada.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace the title stored at books[0] in a two-object Book array.

Index 0 refers to the first object; its Title is Ada.



IDO · WORKED STATE: 0 → Ada, 1 → Grace

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

In a different array, which title prints after updating the object at index 1?
Source code / input:

```
Book[] b={new Book{Title="One"},new Book{Title="Two"}};
b[1].Title="Updated";
Console.WriteLine(b[1].Title);
```

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Given shelf titles Dune,Holes, give output of shelf[1].Title and reason. [4]

Question 2

Write C# to create an array of two Book objects with titles A and B, then print first title. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Two array elements reference the SAME Book object. Changing its title via first reference is then printed via second. Explain. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Updating a property through one reference can be visible through another alias.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

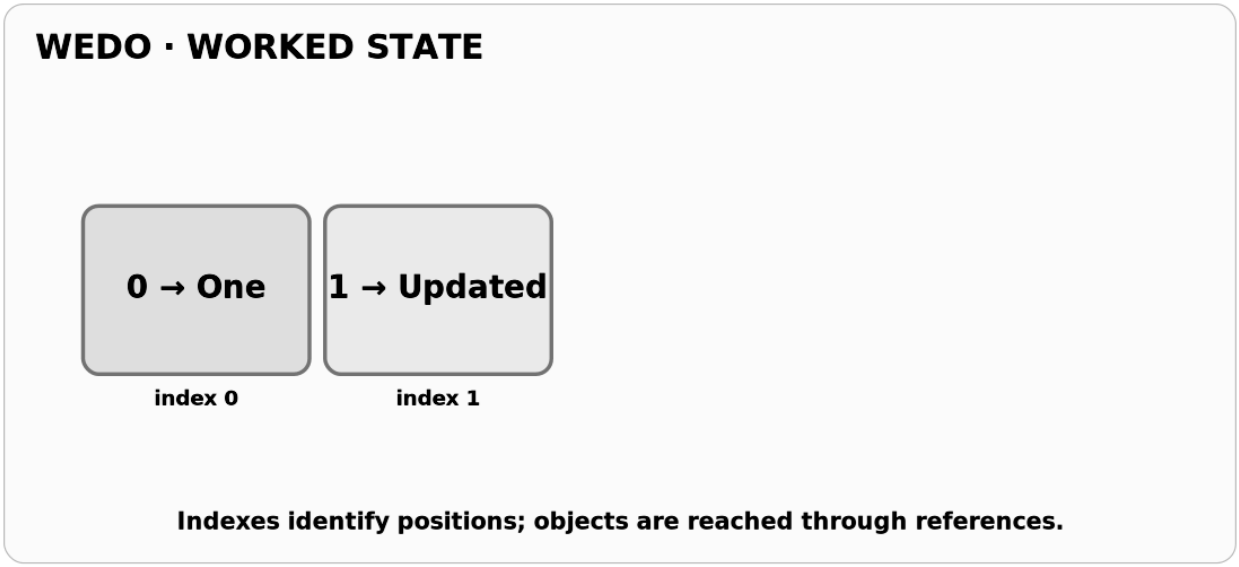
Trace and justify
Index 1 refers to the second object. The property update changes its Title; output is Updated.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

In a different array, which title prints after updating the object at index 1?

Index 1 refers to the second object. The property update changes its Title; output is Updated.



WEDO · WORKED STATE: 0 → One, 1 → Updated